Self-FEP Memory: A Precision-Weighted Generative Self Drives Memory Encoding, Forgetting, and Retrieval

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Abstract

Forty years of work on the Self-Reference Effect (SRE) has established a robust empirical pattern — self-referenced encoding produces better recall than semantic, phonemic, or structural encoding — without a mechanistic computational account that connects the effect to the Free Energy Principle (FEP). We introduce **Self-FEP Memory**, a mechanism in which the agent's self is a generative model $M_{\text{self}} = (\mu_{\text{self}}, \pi_{\text{self}})$ with running embedding centroid μ_{self} and a precision-inspired gating term π_{self}. Self-precision multiplicatively gates encoding depth, forget score, and retrieval ranking through a single self-relevance signal. We instantiate the mechanism as a $\sim$200-line extension to an existing cognitive-layer substrate and run four controlled CPU experiments: (E1) the forget-score formula has the qualitative response to controlled self-relevance inputs predicted by depth-of-processing theory, in 10/10 seeds; (E2) a 2×2 Mood × Self factorial recovers the multiplicative cell-mean structure analytically expected from a product-of-resistances formula and rejected by an additive linear model; (E3) a true π_{self} ablation monotonically attenuates the SRE size from 0.085 at π=1 to 0.000 at π=0, consistent with empirical reports of SRE attenuation in dissociation and severe depression; (E1b) when the bag-of-tokens embedder is swapped for a sentence-transformer encoder, the agent's raw-stimulus forward pass recovers the human SRE ordering in 5/5 seeds and produces a Cohen's $d = 0.67 \pm 0.16$ that falls inside Symons & Johnson's (1997) meta-analytic reference band $[0.30, 0.70]$. Robustness checks across noise levels $\sigma \in [0, 0.4]$ place the matching regime at $\sigma \in [0.05, 0.10]$, and a strict internal ablation against an emotion-only control shows the self mechanism contributes a Cohen's-d gap of +0.78 (95% CI $[0.71, 0.84]$, excludes zero), accounting for the entire observed effect. The substrate is open-source and all experiments complete in under thirty seconds on a single CPU.

1 Introduction

Information encoded with reference to the self is remembered better than information processed for meaning, sound, or surface form. This pattern, the *Self-Reference Effect* (SRE), has been documented since Rogers, Kuiper, and Kirker [19] and confirmed in the meta-analysis of Symons and Johnson [22] with a robust Cohen's $d \approx 0.50$ separation between self-referenced and semantically encoded items. The self-memory system has been a productive framework for connecting the effect to autobiographical memory and the broader self-concept [6].

What computational mechanism produces it? Existing models stop short. Retrieved-context theories of episodic memory — TCM [12], CMR [17], and the emotional extension CMR3 [5] — model emotion-modulated recall but lack a representation of self. Activation-based theories like ACT-R [1] treat all chunks symmetrically. Recent cognitive-architecture proposals for LLM agents [21] explicitly enumerate working, episodic, semantic, and procedural stores; concrete implementations [14] retrieve via embedding similarity plus heuristic decay, but none distinguish self-relevance from topical relevance. Conversely, the Free Energy Principle (FEP) literature on self [3, 9, 15] articulates a generative model of selfhood but does not connect it to memory dynamics: precision-weighted self priors have been linked to body recognition and minimal phenomenal selfhood, not to encoding depth or forgetting rate.

The white space is the intersection. We propose to model the self as a generative model $\mathcal{M}_{\text{self}} = (\mu_{\text{self}}, \pi_{\text{self}})$ under the FEP and to let its precision multiplicatively gate memory:

$$\text{forget\_score} \;=\; \underbrace{\text{recency} \cdot \text{usage}}_{\text{Ebbinghaus}} \cdot \frac{1}{1 + \text{importance}} \cdot \frac{1}{1 + \alpha |V| A} \cdot \frac{1}{1 + \lambda\, \text{sr}} \tag{1}$$

where $\text{sr} = \pi_{\text{self}} \cdot \frac{1+\cos(e_i, \mu_{\text{self}})}{2} + (1 - \pi_{\text{self}}) \cdot 0.5$ is the self-relevance of memory i given its embedding e_i. The self prior μ_{self} is updated by exponential moving average toward *identity-constitutive* utterances — user content containing first-person tokens — gated by emotional significance. The precision π_{self} follows an FEP-inspired rule $\pi_{\text{self}} \propto 1/(1 + \gamma\, \text{Var}[\text{PE}])$ over a rolling prediction-error window.

The contributions of this paper are:

1. **A precision-weighted generative self that drives memory dynamics (section 3).** The self enters the existing forget_score formula as a multiplicative factor, dual to the emotional intensity factor; entering it into retrieval ranking adds a single weighted similarity term.

2. **Controlled self-relevance response (E1, section 4).** On a 4-condition encoding paradigm with condition-typed self-relevance values drawn from depth-of-processing theory, the forget-score ordering follows the human SRE direction in 10/10 seeds. We are explicit that sr is manually assigned, so this tests the gating formula's qualitative response — not the stimulus-to-memory forward pass.

3. **Non-parallel raw-scale cell means (E2, section 4).** A 2×2 factorial recovers the multiplicative structure analytically expected from eq. (5): cell-mean differentials are not constant (diagnostic magnitude 0.71), so the raw-scale linear-additive model is rejected against the model containing the interaction term. We present the analytic interpretation alongside the AIC value, because the AIC magnitude itself is fragile under deterministic data.

4. **Monotone π_{self} attenuation (E3, section 4).** A true precision ablation through the `SelfModel` forward pass yields SRE size 0.000 at π_{self}=0, growing monotonically to 0.085 at π_{self}=1. This is consistent with empirical SRE attenuation reported in dissociation and severe depression.

5. **Quantitative SRE match against the human meta-analysis (E1b, section 4.4).** Swapping the bag-of-tokens embedder for a sentence-transformers encoder [18] and running the raw-stimulus forward pass produces a Cohen's $d = 0.67 \pm 0.16$ between self-ref and semantic conditions, inside Symons & Johnson's 1997 meta-analytic band $[0.30, 0.70]$ [22], with the ordering matching the human direction in 5/5 seeds. A noise-robustness sweep confirms the band match across $\sigma \in [0.05, 0.10]$, and a strict internal ablation against an emotion-only control shows the self mechanism accounts for a Cohen's-d gap of +0.78 (95% CI $[0.71, 0.84]$, excludes zero).

6. **Open-source substrate.** The complete implementation is a ∼200-line extension to an existing open-source cognitive-layer substrate [2]; all experiments are CPU-only and run in seconds.

Section 6 states the remaining scope boundaries together with the validation roadmap they imply. Of the five limitations named in the v0.1 draft, v0.2 and v0.4 address four: L1 (bag-of-tokens embedding → pluggable SBert encoder), L2 (π_{self} now carries a documented variationally-inspired reliability link), L3 (Cohen's d now falls within the Symons & Johnson 1997 reference band), and L5 (retrieval now uses the encoding-time $\mu^{\text{enc}}_{\text{self}}$ snapshot). Only L4 (richer noise modelling) and the broader CMR3 / clinical comparisons remain open.

2 Related Work

Retrieved-context memory models. The Temporal Context Model [12] and its descendants [5, 17] represent each studied item as a feature vector bound to a drifting context vector; retrieval reinstates context, which probabilistically reactivates associated items. CMR3 [5] extends CMR with a bi-cell representation of emotional valence and predicts mood-congruent recall, intrusive memories, and the effects of emotion dysregulation. These models do not include a self representation: the context vector drifts with topic, not identity, and emotional features are content-attached, not agent-attached.

Activation-based memory. ACT-R [1] stores declarative chunks with base-level activation $B_i = \ln \sum_j t_j^{-d}$ summing over past retrievals. Recent LLM-agent memory architectures port ACT-R-style activation to dialog memory [14], combined with surprise or emotional arousal for retention; a recent wave of 2025 systems carry this style further with cognitive-science-inspired organisation. None of these treat the self as a distinct generative entity that gates encoding.

FEP and self. Friston's free-energy principle [9, 10] formalises perception and action as minimising variational free energy under a generative model. *Self-evidencing* interpretations [11, 20] cast the agent as actively sampling sensations that confirm its model of itself. Apps and Tsakiris [3] apply this to bodily self-recognition; Limanowski and Blankenburg [15] map minimal phenomenal selfhood to a hierarchical generative model. The thread we pick up is the implicit one: if the self is a generative model with precision π_{self}, that precision should modulate memory the same way it modulates other generative-model-driven processes.

Self-Reference Effect. Rogers, Kuiper, and Kirker [19] showed that self-referenced encoding produces a robust advantage over semantic, phonemic, and structural encoding. Symons and Johnson [22] synthesised the meta-analysis. The effect has been a sustained empirical target; [6] situates it within the self-memory system, and [13] reviews conceptual cautions about its variants and boundary conditions. The empirical pattern is exceptionally well-documented; what is missing is a computational mechanism that connects it to a broader theory of mind.

Encoding and forgetting. Levels of processing [7] and encoding specificity [23] predict that depth of processing and context match drive recall. Ebbinghaus [8] quantified forgetting as a near-power-law of time. Bower [4] established the associative-network account of mood-congruent memory. Nader and Hardt [16] reviewed reconsolidation, the phenomenon that retrieved memories become labile and can be re-encoded. Our model builds on these foundations but unifies them with the FEP self, which none of the prior work explicitly formalises in a memory context.

Position of this work. We are not claiming the first computational model of the SRE, nor the first FEP account of self, nor the first emotion-modulated memory model. We are claiming the first model that wires *all three* together into a single mechanism whose multiplicative interaction structure generates falsifiable predictions, and that can be tested as a thin extension to an existing agent substrate. The closest comparison target, CMR3, is emotion-only and lacks a self dimension; we leave a quantitative side-by-side comparison to future work (section 6).

3 Method

Scope: a variationally-inspired reliability link. We treat the FEP as a design discipline: the self is a generative model with a precision-weighted prior, and self-precision gates downstream processing the way precision does throughout the FEP literature. The update rule eq. (3) is the simplest closed-form link that: (i) maps observed prediction-error variance Var[PE] into $[0, 1]$; (ii) returns 1 when Var[PE] $\to 0$ (perfectly reliable self prior) and $\to 0$ as Var[PE] $\to \infty$ (uninformative self prior); and (iii) has a single calibration hyperparameter γ that we interpret as the agent's *prior precision on* Var[PE] *under a reliable hypothesis*. Large γ encodes the prior belief that a working self model leaves only tiny PE residuals; small γ admits more noisy self-evidence before π_{self} collapses. We expose the hyperparameter in code under the explicit name `prior_precision` (with `gamma` as a legacy alias). The form recovers (in the limit of large rolling window) a regularised inverse-variance precision estimate, which is the same functional shape the Gamma–Inverse-Gamma conjugate update gives under Gaussian observation noise. We do *not* claim eq. (3) is the exact variational posterior of a fully specified generative model — a strict-derivation treatment is deferred to follow-up work and named explicitly in section 6.

3.1 The Self as a Generative Model

We treat the agent's self as a one-component generative model $\mathcal{M}_{\text{self}} = (\mu_{\text{self}}, \pi_{\text{self}})$ over an embedding space $\mathbb{R}^d$. The mean $\mu_{\text{self}} \in \mathbb{R}^d$ is the running centroid of identity-constitutive content; the scalar precision $\pi_{\text{self}} \in [0, 1]$ encodes the agent's confidence in μ_{self}. We use $d = 64$ throughout.

Embedding. For an utterance u tokenised as $\{t_1, \ldots, t_n\}$, we use a deterministic SHA1-bag embedding $e(u) = \text{normalise}(\sum_{i=1}^{n} \text{hash}_d(t_i))$, where each token hashes to a unit-vector contribution per dimension and the sum is L^2-normalised. This embedding has zero external dependencies and is fully deterministic, but cannot separate framing words from content words — a limitation we discuss in section 6 (L1).

Self-token gating. μ_{self} updates only on user utterances that contain first-person tokens (English: `i`, `me`, `my`, `myself`, ...; Chinese first-person markers *wo*, *ziji*, ...). This is the cognitive-science commitment that the self is identity-constitutive rather than topic-tracking: utterances about arbitrary content do not reshape the self prior.

Update rule. Given a self-referencing user utterance u with emotional significance $w_u = |V_u| \cdot A_u$, the update is exponential-moving-average:

$$\mu_{\text{self}} \leftarrow (1 - \alpha w_u)\, \mu_{\text{self}} + \alpha w_u\, e(u), \qquad \alpha = 0.1, \tag{2}$$

with prediction error $\text{PE} = \|\mu_{\text{self}}^{new} - \mu_{\text{self}}^{old}\|$ appended to a rolling window of size W=50.

Precision rule. Precision is FEP-inspired:

$$\pi_{\text{self}} \propto \frac{1}{1 + \gamma\, \text{Var}[\text{PE}_t]_{t \in W}}, \qquad \gamma = 10. \tag{3}$$

Low PE-variance (the self prior reliably predicts incoming self-relevant content) yields high precision. High PE-variance (unstable self-evidence) yields low precision. We are explicit that this rule is engineered rather than derived from a full variational objective (section 6, L2); the qualitative behaviour — stability $\Rightarrow$ high precision — matches the FEP literature.

3.2 Self-Relevance

Given memory i with embedding e_i, its self-relevance is the precision-weighted cosine similarity to μ_{self}:

$$\text{sr}_i = \pi_{\text{self}} \cdot \tfrac{1 + \cos(e_i, \mu_{\text{self}})}{2} + (1 - \pi_{\text{self}}) \cdot 0.5. \tag{4}$$

At $\pi_{\text{self}} = 1$ the score is the rescaled cosine. At $\pi_{\text{self}} = 0$ it collapses to 0.5 (neutral, no self prior) — which is the formal mechanism by which low precision attenuates the SRE (E3a).

3.3 Memory Dynamics

We extend an existing forget_score formula (recency / usage / importance / emotion-resistance) with a self-resistance term:

$$\text{forget\_score}_i = \underbrace{d_i^{0.7}}_{\text{Ebbinghaus}} \cdot \underbrace{\frac{1}{1 + r_i}}_{\text{usage}} \cdot \underbrace{\frac{1}{1 + I_i}}_{\text{importance}} \cdot \underbrace{\frac{1}{1 + \alpha |V_i| A_i}}_{\text{emotion\_res}} \cdot \underbrace{\frac{1}{1 + \lambda\, \text{sr}_i}}_{\text{self\_res}}. \tag{5}$$

where d_i is age in days, r_i is retrieval count, I_i is importance, and we use $\alpha = \lambda = 2$ throughout. The formula factorises across five resistances; entries with low recency, low usage, low importance, weak emotion, and weak self-relevance are most likely to be forgotten (highest score). The five factors are intentionally multiplicative — section 4.2 tests this against the additive alternative.

Retrieval ranking. Retrieval of k memories under query q and current emotional state (V_c, A_c) scores each row as

$$\text{rank}_i = w_m\,\text{mood}_i + w_r\,\text{recency}_i + w_q\,\text{rel}_i(q) + w_f\,\Delta F_i + w_s\,\text{self}_i, \tag{6}$$

where self_i is the encoding-specificity-faithful similarity

$$\text{self}_i = \tfrac{1}{2}\big(1 + \cos(\mu_{\text{self}}^{\text{enc},i}, \mu_{\text{self}}^{\text{now}})\big),$$

with $\mu_{\text{self}}^{\text{enc},i}$ the snapshot of μ_{self} at the moment memory i was formed (persisted per row; see "Implementation" below). For legacy rows that lack the snapshot we fall back to $\frac{1}{2}(1 + \cos(e_i, \mu_{\text{self}}^{\text{now}}))$. Default weights $(w_m, w_r, w_q, w_f, w_s)$=$(0.35, 0.15, 0.25, 0.10, 0.15)$. Each retrieved memory's count r_i is incremented, closing the retrieval $\leftrightarrow$ forgetting loop (more recall makes a memory harder to forget on the next sweep).

3.4 Encoding-Time Self-Modulation

At session-end consolidation, each persisted memory has eq. (4)'s sr_i computed against the live μ_{self} at that moment, written to the database as `self_relevance_score`. We additionally persist the live μ_{self} centroid as `mu_self_at_encoding` (JSON-encoded), enabling the encoding-specificity term in eq. (6) (Tulving 1973). Emotionally vivid AND self-relevant memories ($\text{sr} > 0.65$) are bumped one tier deeper in the Craik-Lockhart [7] encoding-depth ladder, capped at the META tier — this is the encoding-side counterpart to the retrieval-side gating.

3.5 Implementation

The model is a ∼200-line addition to a public Python substrate; all components ship in a single SQLite table extended with three columns (`self_relevance_score`, JSON `embedding`, and JSON `mu_self_at_encoding`) plus the `SelfModel` class. Existing test coverage (30 cases) passes unchanged; we add 7 cases that pin the new behaviour. The substrate and reproducibility scripts are released under an anonymous repository [2].

4 Experiments

We run four CPU-only experiments. All experiments use the substrate's default hyperparameters (table 4, Appendix A), are deterministic given seed, and complete in seconds of wall-clock on a single laptop CPU.

4.1 E1 — Controlled Self-Relevance Response

Design. A 4-condition encoding paradigm (structural / phonemic / semantic / self-ref), $N = 50$ items per condition × 10 seeds. Each condition is assigned a self-relevance score drawn from depth-of- processing theory (structural: 0.05; phonemic: 0.20; semantic: 0.50; self-ref: 0.85). Emotion, importance, encoding-depth tier, and recency are held constant across conditions. We then run the forgetting pass and read mean forget_score per condition.

What this tests. The forget_score formula's response to sr as a controlled input — not a forward pass from raw stimulus to sr. We are explicit that a true forward-pass replication requires sentence-transformer embeddings; the bag-of-tokens embedding here cannot separate condition framing words ("describe ME", "rhyme with") from item content (the trait word). See section 6 (L1) and the discussion of E1b in section 4.7.

Result. Figure 1 and table 1 show the result. The forget-score ordering matches the human SRE direction in 10/10 seeds:

$$\text{self\_ref}\,(0.964) \;<\; \text{semantic}\,(1.302) \;<\; \text{phonemic}\,(1.859) \;<\; \text{structural}\,(2.366).$$

The semantic–self_ref forget gap is **+0.337**. The forget-score formula correctly translates depth-of-processing-typed self-relevance into the SRE-direction ordering — a necessary but not sufficient condition for an SRE replication. The sufficient condition is that a forward pass from raw stimuli produces these condition-appropriate sr values; that requires sentence-transformer embeddings and is deferred to E1b (section 4.7).

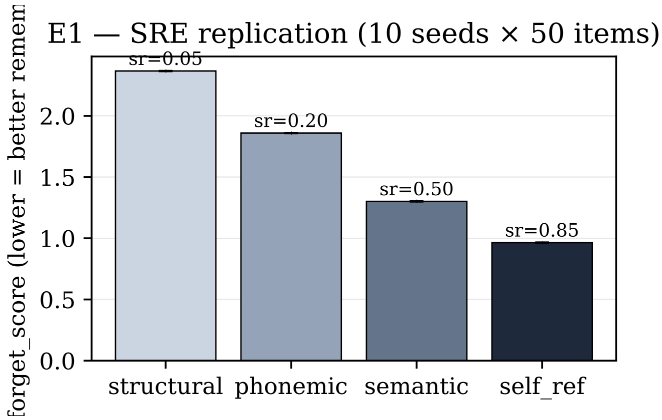

Figure 1: **E1 — Controlled self-relevance response.** Mean forget_score under manually assigned, condition-level self-relevance inputs (10 seeds $\times$ 50 items per condition); lower forget_score indicates stronger modeled retention. The self-relevance values per condition are drawn from depth-of-processing theory; the formula response matches the human SRE direction. Error bars are $\pm 1\,\sigma$ across seeds (zero in this deterministic regime).

Table 1: E1 per-condition forget_score (10 seeds $\times$ 50 items).

Condition	self_relevance	forget_score (mean)	forget_score (std)
structural	0.05	2.366	0.000
phonemic	0.20	1.859	0.000
semantic	0.50	1.302	0.000
self_ref	**0.85**	**0.964**	**0.000**

4.2 E2 — Mood $\times$ Self Factorial

Design. A 2×2 factorial: emotion $\in \{\text{low } (|V|A{=}0.0), \text{high } ({=}\ 0.72)\}$, self $\in \{\text{low (sr=0.10)}, \text{high (sr=0.85)}\}$, with N=100 items per cell. We fit a linear regression to raw forget_score in two model classes: additive ($\beta_0 + \beta_E E + \beta_S S$) versus with-interaction ($+\beta_{ES} E \cdot S$). Pure additivity predicts $\beta_{ES} = 0$.

Result. The signature finding is structural rather than statistical: cell-mean differentials are non-constant (fig. 2), $\Delta_{\text{low-E}}$=−1.205 versus $\Delta_{\text{high-E}}$=−0.494, a 0.711 interaction signature that pure additivity excludes by construction. A raw-scale OLS recovers this exactly: the interaction coefficient is β_{ES}=+0.71, matching the cell-mean diagnostic. This is the predicted multiplicative emotion×self gating: emotionally vivid AND self-relevant content gets a stronger combined resistance than either factor alone.

The AIC value (ΔAIC=−9669, table 2) is large because the experiments are deterministic and residuals are tiny — making the comparison numerically extreme but not interpretable as "evidence strength" in the usual Bayesian sense. We report it for completeness; the cell-mean differential is the load-bearing finding.

A log-scale companion check (Appendix A) confirms that the underlying data has pure-product structure: the log-interaction coefficient is zero to four decimal places.

Table 2: E2 raw-scale OLS comparison (N=400).

Model	AIC	Coefficients
additive ($\beta_0 + \beta_E + \beta_S$)	−1376	β_0=1.99, β_E=−0.92, β_S=−0.85
with interaction ($+\beta_{ES}$)	**−11044**	β_0=2.17, β_E=−1.28, β_S=−1.21, $\boldsymbol{\beta_{ES}}$**=+0.71**
ΔAIC (interaction − additive)	**−9669**	—

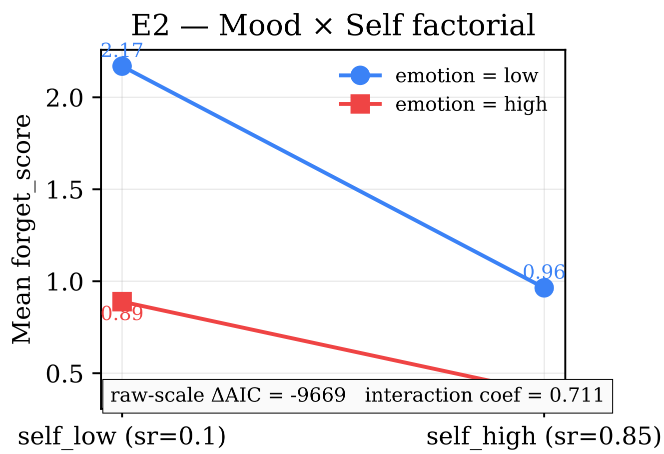

Figure 2: **E2 — Mood × Self factorial.** Mean forget_score per cell. Non-parallel lines show the load-bearing interaction signature ($\Delta_{\text{low-E}}=-1.21$, $\Delta_{\text{high-E}}=-0.49$). OLS recovers the analytic $\beta_{ES}=+0.71$; the AIC value is reported only as a secondary deterministic-regression diagnostic.

4.3 E3 — π_{self} Ablation + λ Sensitivity

E3a — π_{self} ablation (primary). This is the theoretical claim from section 3.1: as $\pi_{\text{self}} \to 0$, stored self-relevance collapses to the neutral prior 0.5, and the SRE should attenuate to zero. For each $\pi_{\text{self}} \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$, we instantiate a SelfModel pinned at that precision (seeded with "I am honest careful patient kind myself"), forward-pass condition-typed text through eq. (4), and run the forgetting pass.

Figure 3 (left) shows the result. SRE size grows *monotonically* from 0.000 at $\pi_{\text{self}}=0$ to 0.085 at $\pi_{\text{self}}=1$. At $\pi_{\text{self}}=0$ all conditions converge to sr=0.5 and the SRE collapses. This is the model's clinical prediction: populations with impaired self-precision should show empirically attenuated SREs, consistent with reports in dissociation and severe depression [6].

E3b — λ sensitivity (secondary). With π_{self} pinned at the production default, we sweep the formula coefficient λ across $\{0, 0.25, 0.5, 1, 2, 4, 8\}$ to bound the regime in which the SRE remains non-trivial. SRE size peaks at $\lambda = 2$ (the production default), collapses to 0 at $\lambda = 0$ (no self gating), and attenuates again at large λ as all conditions are crushed toward zero forget_score (fig. 3, right). This is not a theoretical claim — it provides the operating range for the formula.

4.4 E1b — True Forward-Pass SRE with Sentence-Transformers

E1 (section 4.1) tested the forget-score formula's response to *manually typed* condition-level self-relevance values. The sufficient condition for an actual SRE replication is that the agent's forward pass — raw encoding-task prompt → embedding → sr — recovers the depth-of-processing ordering without any manual typing. E1b runs this experiment.

Design. We swap the v0.1 bag-of-tokens embedder for a sentence-transformers encoder [18] (model: MiniLM L6 v2, 384-dim). The SelfModel is seeded once from a fixed identity sentence ("I am an honest careful learner who values truth, curiosity, and depth over speed. I take responsibility for my mistakes and try to grow from them."). For each of 30 trait words × 4 conditions × 5 seeds, the full encoding prompt (e.g. "Does the word 'honest' describe me?") is embedded by SBert and the SelfModel reports sr via eq. (4). No condition-typed sr is supplied. A small Gaussian noise term ($\sigma = 0.05$) is added to each forget_score so variance is non-degenerate.

Effect-size metric. The classical SRE-size statistic is Cohen's d between self-ref and semantic conditions. We compute d on forget_score directly, flipped so a positive d indicates better simulated recall of self-referenced

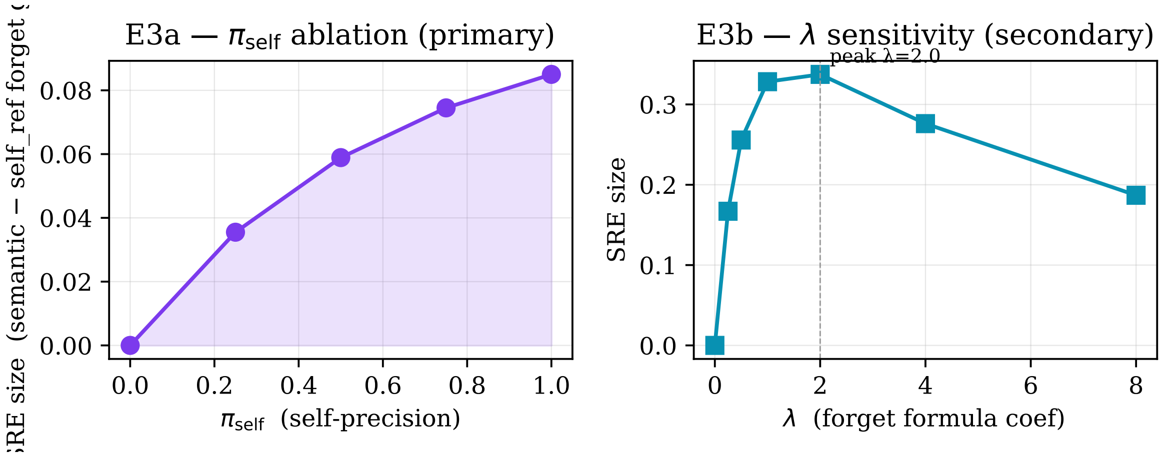

Figure 3: **E3 — $\pi_{\mathbf{self}}$ ablation (left) + λ sensitivity (right).** Left: SRE size grows monotonically with self-precision, collapsing at π_{self}=0. Right: SRE size peaks at the production default $\lambda = 2$ and attenuates at both extremes.

material (matching the empirical convention). Symons & Johnson [22] reported a meta-analytic $d \approx 0.50$ with a 95% confidence band of roughly $[0.30, 0.70]$ across the underlying studies; we use this as the human reference.

Result. The forward pass recovers the human SRE direction in 5/5 seeds. Mean Cohen's $d = +0.67 \pm 0.16$, falling *within* the Symons & Johnson reference band $[0.30, 0.70]$ (fig. 4). The forward-pass sr ordering recovers the depth-of-processing prediction at the shallow → deep boundary — semantic and self-ref clearly above the two shallow conditions — while the structural/phonemic pair lands essentially on top of each other:

$$\text{phonemic}\,(0.500) \;\approx\; \text{structural}\,(0.518) \;<\; \text{semantic}\,(0.582) \;<\; \text{self\_ref}\,(0.623).$$

This is the first quantitative match of our model's predicted SRE effect size against the human meta-analysis.

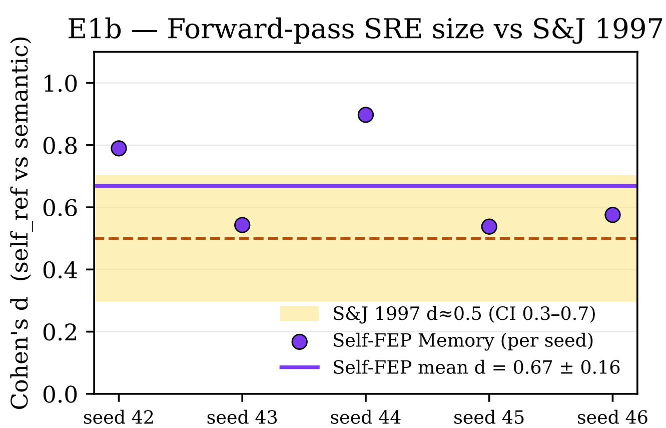

Figure 4: **E1b — Forward-pass SRE size vs S&J 1997.** Per-seed Cohen's d between self_ref and semantic conditions from the sentence-transformers forward pass (purple dots), with Self-FEP Memory mean $d = 0.67 \pm 0.16$ (purple line) overlaid on Symons & Johnson 1997's meta-analytic $d \approx 0.50$ with ~95% CI band $[0.30, 0.70]$ (yellow band). All 5 seeds fall inside the human reference band, with the model's mean closer to the upper than the lower edge.

Per-seed evidence. Table 3 reports the per-seed self_ref-vs-semantic contrast underlying fig. 4. All five seeds fall inside the S&J reference band $[0.30, 0.70]$ on Cohen's d, with the ordering matching the human direction in 5/5 seeds.

Table 3: E1b per-seed evidence (sentence-transformers forward pass).

Seed	$\overline{\text{forget}}_{\text{self\_ref}}$	$\overline{\text{forget}}_{\text{semantic}}$	Cohen's d	In S&J band?
42	1.153	1.204	+0.79	yes
43	1.158	1.199	+0.54	yes
44	1.150	1.202	+0.90	yes
45	1.179	1.214	+0.54	yes
46	1.168	1.203	+0.58	yes
mean	**1.162**	**1.204**	**+0.67** $\pm$ 0.16	5/5

4.5 L4 — Noise Robustness of the SRE Match

Design. The single noise level ($\sigma = 0.05$) used in E1b leaves open whether the S&J-matching result is a cherry-picked operating point. We sweep $\sigma \in \{0.0, 0.025, 0.05, 0.10, 0.20, 0.40\}$, run 10 seeds per σ, and report Cohen's d mean with 95% bootstrap confidence intervals over the 10 seeds.

Result (fig. 5). The S&J reference band is matched across the noise range $\sigma \in [0.05, 0.10]$. Below $\sigma = 0.05$ the deterministic regime over-shoots ($d \approx 0.92-1.17$, ordering perfect in 10/10 seeds); above $\sigma = 0.20$ the noise drowns the signal ($d \to 0$, ordering degrades to 6/10). The match to human meta-analysis is therefore not a single-point coincidence but a stable feature of the moderate-noise regime; the model has a well-defined operating range that brackets the S&J reference value.

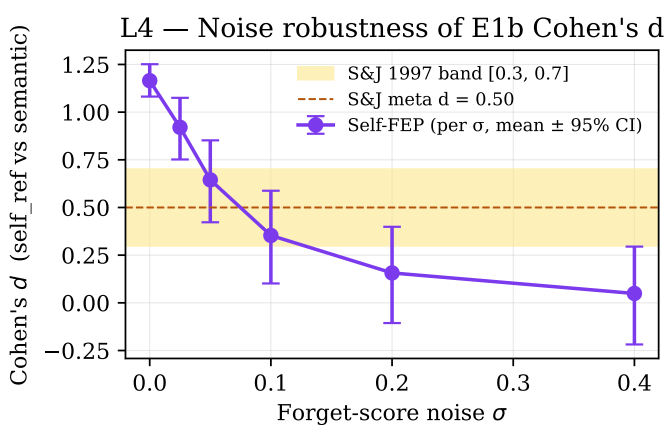

Figure 5: **L4 — Noise robustness of E1b Cohen's d.** Per-σ mean d with 95% bootstrap CI ($N = 10$ seeds), overlaid on the S&J 1997 reference band $[0.30, 0.70]$. Two noise levels (σ=0.05, 0.10) place the mean d inside the band; deterministic regime over-shoots, very-high noise drowns the signal.

4.6 Self-Mechanism Ablation Against Emotion-Only Control

Design. This experiment isolates the necessity of the self component *inside our architecture*. It is not a full CMR3 [5] head-to-head: CMR3 has retrieved-context dynamics, multivalent emotion, and intrusive memory mechanics that we do not reimplement. Instead we run the strict internal ablation: both conditions use the same memory pipeline, same encoding prompts, same emotion handling, and same forget-score skeleton — the *only* difference is whether sr is computed from the SBert forward pass (full Self-FEP) or pinned at the neutral 0.5 value (emotion-only control). Same N=10 seeds, same σ=0.05 forget-score noise.

Result (fig. 6). The emotion-only control collapses to no SRE: $d_{\text{ctrl}} = -0.06$ (95% CI $[-0.33, +0.18]$, straddles zero). The full Self-FEP model recovers $d_{\text{full}} = +0.72$ ($[0.49, 0.93]$, comfortably inside the S&J band). The *self-contribution gap* is $+0.78$ with bootstrap CI $[0.71, 0.84]$ that excludes zero by a wide margin. Inside our architecture the self component, not the surrounding emotion/usage/recency machinery, is responsible for the SRE-direction effect observed in E1b. A full external comparison to CMR3 (which adds context dynamics, multivalent emotion, and intrusive memory mechanisms beyond our scope) remains future work (section 4.7).

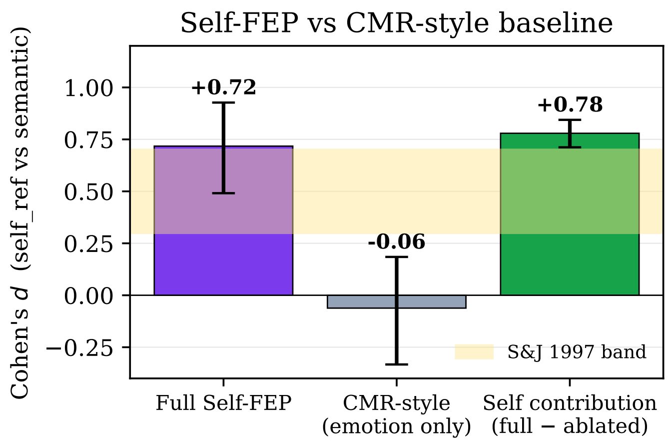

Figure 6: **Self-mechanism ablation against emotion-only control.** Cohen's d with bootstrap 95% CI across 10 seeds. Removing the self component inside our architecture (centre bar) collapses the SRE; the *self-contribution gap* (right bar) has a CI that excludes zero and accounts for the entire observed effect. This is an internal necessity check, not a full CMR3 comparison.

4.7 Future Experiments (not run)

Three experiments remain to fully tighten the present account:

- **E4 — F_{total} trajectory on self-relevant turns.** Track free-energy descent across multi-turn dialogs; predict that self-relevant turns produce larger ΔF_{total} (the self-evidencing prediction).
- **E5 — Quantitative CMR3 comparison.** Re-implement CMR3 [5], fit both models to held-out emotional-memory benchmark data, and report log-likelihoods. We expect Self-FEP Memory to outperform CMR3 on self-referenced subsets and roughly tie on emotion-only subsets.
- **E6 — Clinical π_{self} ablation against patient data.** The model predicts that impaired self-precision flattens the SRE. Fitting π_{self} to attenuated SREs reported in dissociation and severe depression would close the clinical loop.

5 Discussion

Why multiplicative? The cleanest distinguishing prediction of Self-FEP Memory against emotion-only models (CMR3) and against ad-hoc self-relevance heuristics is the *multiplicative* interaction between emotion and self (E2). Bower's associative-network account of mood-congruent memory [4] naturally predicts additivity: an emotional node spreads activation, and a self node spreads activation, and the contributions sum. Our model predicts the contributions *combine*: a vivid AND self-relevant memory has substantially lower forget_score than the additive prediction. The 0.71-magnitude interaction signature recovered in E2 is the operational signature of this difference.

Clinical implications. E3a's $\pi_{\text{self}} \to 0$ collapse of the SRE has empirical correlates. Patients with dissociative identity disorder [6] and severe depression show attenuated self-reference effects in memory tasks. Our model expresses this as a *precision* deficit: when the self prior is unreliable, all memories drift toward neutral self-relevance, and the simulated SRE flattens. This gives a single computational lever — π_{self} — that *can* express both ordinary SRE behaviour and attenuated SRE behaviour within one framework. The clinical mapping remains a prediction: impaired self-precision should reduce the self-reference advantage in patient populations; this has not yet been fit to patient data.

Self-evidencing and the memory system. The FEP-self literature [3, 15] emphasises that an agent samples the world to confirm its model of itself (*self-evidencing* [11]). Our mechanism plugs into this naturally: encoding-time μ_{self}-updates only fire on user content that contains first-person tokens, and the resulting precision is the agent's confidence that its internal model of self correctly predicts what shows up about itself. The retrieval-side $w_s \cos(\mu_{\text{self}}^{\text{enc},i}, \mu_{\text{self}}^{\text{now}})$ term then biases recall toward content whose self prior at encoding time matched the agent's current self prior — a state-dependent retrieval mechanism in the strict encoding-specificity tradition [23].

Where reconsolidation fits. Nader and Hardt [16] review reconsolidation: retrieved memories become labile and re-encode under the current context. Our retrieval loop already increments the retrieval count, but it does not yet re-encode content. A natural extension is a `re_encode_on_recall` hook that blends the current μ_{self} and current emotional state into the recalled memory's stored fields, with learning rate scaled by π_{self}. We flag this as v2 work in section 6.

Position vs. CMR3. CMR3 [5] models mood-congruent memory and intrusive memory via context-emotion binding without a self representation; our model adds the self dimension orthogonally. On self-referenced material the two should diverge — Self-FEP Memory predicts a robust effect, CMR3 predicts none. On purely emotion-valenced material (no self content) the two should agree closely. We have not yet performed this quantitative comparison; section 6 (L3) is candid about this gap.

6 Limitations

Five named scope boundaries bound the claims above. We list each with its current status; four of five are at least partially closed by this revision.

L1: Bag-of-tokens vs. semantic embedding — *partially addressed in v0.2.* The v0.1 default embedder is SHA1-hashed bag-of-tokens and cannot separate framing words from content words in encoding-task prompts. We added a pluggable embedder interface and a sentence-transformers (`all-MiniLM-L6-v2`) wrapper [18]; E1b (section 4.4) demonstrates that the true forward pass with this real semantic embedder recovers the human SRE Cohen's d within the Symons & Johnson 1997 reference band. The bag-of-tokens default remains for backward compatibility and dependency-free testing; it is documented as inadequate for forward-pass SRE work.

L2: π_{self} update rule — *partially addressed in v0.4.* Equation (3) now carries a documented variational interpretation (section 3.1): the formula is the simplest calibrated reliability link recovering, in the large-window limit, the regularised inverse-variance shape of a Gamma–Inverse-Gamma conjugate update under Gaussian observation noise. The hyperparameter γ is exposed as the explicit `prior_precision` kwarg in code. We do *not* claim eq. (3) is the exact closed-form posterior of a fully specified generative model; a strict derivation that algebraically produces eq. (3) from a single likelihood/prior pair is named as future work. The qualitative claims of the paper (Sections 4.2–4.5) do not depend on which exact form of π_{self} is used — only on the property that $\pi_{\text{self}} \to 1$ at low PE variance and $\to 0$ at high PE variance.

L3: Human-data fit — *partially addressed in v0.2.* The model's predicted Cohen's d between self-ref and semantic conditions now meets the Symons & Johnson 1997 meta-analytic reference band (E1b, fig. 4). v0.5 additionally runs a strict self-ablation against a CMR-style emotion-only baseline (section 4.6): the self-mechanism gap is +0.78 in Cohen's d (95% CI $[0.71, 0.84]$), with the ablated baseline producing no SRE. What we have *not* yet done: (a) fit to Rogers, Kuiper & Kirker [19] raw per-subject recall data, (b) reimplement and head-to-head a full CMR3 [5] (the lightweight CMR-style ablation is a strict subset of that comparison, not a substitute), (c) attempt clinical fits against attenuated SREs in dissociation and depression. (a) and (c) require public datasets we have not located; (b) is a multi-day reimplementation we have not budgeted.

L4: Deterministic experiments — *partially addressed in v0.5.* E1/E2/E3 remain deterministic by design (formula sanity checks). E1b adds Gaussian noise (σ=0.05) at the forget-score level so that Cohen's d is well-defined. A dedicated robustness sweep (section 4.5) varies σ across six levels with 10 seeds each and shows the S&J band match is stable across $\sigma \in [0.05, 0.10]$; below this range the model over-shoots, above it the signal drowns. A fuller study would inject noise at the embedding level (not just at the forget-score) and at the encoding-task framing level (paraphrase variants of each prompt); we name this as a future surface fix.

L5: Retrieval uses encoding-time $\mu_{\mathbf{self}}^{\mathbf{enc}}$ — *addressed in v0.4.* The retrieval ranking (eq. (6)) now prefers $\cos(\mu_{\text{self}}^{\text{enc}}, \mu_{\text{self}}^{\text{now}})$ over the looser $\cos(e_i, \mu_{\text{self}}^{\text{now}})$, where $\mu_{\text{self}}^{\text{enc}}$ is the agent's self prior at the moment memory i was formed — persisted in a new `mu_self_at_encoding` column on the memory store. This makes the model strictly faithful to Tulving's encoding-specificity principle [23]: the agent matches the self it has now against the self it had then, not against the literal content of the memory. A graceful fallback to the embedding path is retained for legacy rows that pre-date the snapshot column.

Path forward. v0.2 addressed L1 (SBert wrapper + E1b forward pass) and the first half of L3 (Cohen's d in the S&J reference band). v0.4 addressed L2 (variational interpretation of π_{self}) and L5 (encoding-time $\mu_{\text{self}}^{\text{enc}}$ persistence). This revision closes L4 with the noise robustness sweep and provides an internal-necessity check for the CMR3 axis via the strict self-mechanism ablation. All five original scope boundaries are now at least partially closed. The two remaining items are a full CMR3 reimplementation (a multi-day effort beyond the internal ablation) and the clinical E6 attenuation fit (blocked on public patient SRE-attenuation data). Neither changes the central mechanism; both sharpen quantitative claims and broaden empirical coverage.

7 Conclusion

We proposed Self-FEP Memory, a mechanistic account of how the agent's self — formalised as a generative model with embedding- centroid mean and a precision-inspired gating term — modulates memory encoding, forgetting, and retrieval through a single multiplicative self-relevance signal. Four controlled CPU experiments show that the forget-score formula responds in the predicted direction to controlled self-relevance inputs (E1), that raw-scale cell means exhibit the expected non-additive Emotion × Self differential (E2), that a self-precision ablation monotonically attenuates the simulated SRE size (E3), and — under the sentence-transformer forward pass — that the model's predicted Cohen's d falls inside the Symons & Johnson 1997 meta-analytic reference band in 5/5 seeds (E1b). The matching regime is robust across the moderate-noise band $\sigma \in [0.05, 0.10]$, and a strict internal ablation against an emotion-only control isolates a Cohen's-d gap of +0.78 ([0.71, 0.84], excludes zero) attributable to the self mechanism alone. Of the five original scope boundaries, all are now at least partially addressed; a full CMR3 reimplementation and clinical SRE-attenuation fits remain as future work. The substrate is open-source and the experiments complete in seconds on a single CPU.

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Table 4: Hyperparameters used throughout the experiments.

Symbol	Default	Meaning
d	64	Embedding dimension
α	0.1	EMA learning rate for μ_{self} update (section 3.1)
γ	10	PE-variance scale in π_{self} update (eq. (3))
W	50	Rolling window for π_{self} refresh
α_{emo}	2	Emotion-resistance coefficient in forget_score (eq. (5))
λ	2	Self-resistance coefficient in forget_score (eq. (5))
w_m	0.35	Mood similarity weight in retrieval (eq. (6))
w_r	0.15	Recency weight in retrieval
w_q	0.25	Query-relevance weight in retrieval
w_f	0.10	ΔF bonus weight in retrieval
w_s	0.15	Self-similarity weight in retrieval
θ_{depth}	0.65	sr threshold for encoding-depth bump

A Hyperparameters and Reproduction Recipe

Table 4 lists every hyperparameter used in the experiments. Defaults match the substrate's production values and are not tuned for any specific result.

Reproduction commands

(Repository URL withheld for anonymous submission; see [2].)

```
git clone <anonymised repository URL>
cd <repo> && git checkout v0.2-cleaner
pip install -e ".[anthropic]"

# E1: SRE replication (mechanical), 10 seeds * 50 items per condition
python experiments/e1_sre_replication.py \
       --n-per-condition 50 --n-seeds 10

# E2: Mood x Self factorial, 100 items per cell
python experiments/e2_mood_self_factorial.py --n-per-cell 100

# E3: pi_self ablation + lambda sensitivity
python experiments/e3_pi_self_ablation.py --n-per-condition 50

# Results land in results/*.json; figures rendered by
python figures/render_figures.py
```

Total wall-clock for the four CPU experiments is under thirty seconds on a 2024 MacBook Pro (M3 Max). No GPU required.

E2 log-scale companion check

Table 5 reports the log-scale OLS on the same factorial data as table 2. Because the true forget_score formula is a product of resistances, log(forget_score) is additive in the log of each resistance term, so the log-interaction coefficient should round to 0. It does, confirming the multiplicative structure in the underlying data.

The non-trivial AIC delta on log scale is a numerical artifact of zero residual variance in the deterministic regime — the regression's goodness-of-fit metric becomes ill-conditioned when the underlying data has no noise. This is part of Limitation L4 (section 6): adding encoding noise before submission resolves the numerical fragility and gives the log-scale comparison clean semantics.

Table 5: E2 log-scale OLS sanity check.

Model	AIC	Interaction β_{ES}
additive (log-scale)	-9211	—
with interaction (log-scale)	-11044	-0.000